



Republic of Rwanda  
Ministry of Education



**RTB** | RWANDA  
TVET BOARD

## DIGITAL ELECTRONICS

**GENDE401**

**Apply Digital Electronics**

### Competence

**RQF Level:** 4

**Learning Hours**

**Credits:** 8



**80**

**Sector:** ICT and Multimedia

**Trade:** Computer System and Architecture

**Module Type:** General

**Curriculum:** ICTCSA4001 TVET Certificate IV in Computer System and  
Architecture

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**1200**

**Issue Date: September, 2023**

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                             |                             |            |                   |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------|-----------------------------|------------|-------------------|
| <b>Purpose statement</b>               | This module describes employable and practicable skills, knowledge and right attitudes required by the learner to apply digital logics, apply Logic ICs, implement Combinational Logic Circuits, and implement Sequential Logic Circuits. The content provided in this course is package reserved for a learner pursuing TVET certificate IV in Computer system and Architecture. Graduates at this level will have knowledge and skills for digital electronics and community involvement and father leaning. |            |                             |                             |            |                   |
| <b>Learning assumed to be in place</b> | ▪ Applied physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                             |                             |            |                   |
| <b>Delivery modality</b>               | <b>Training delivery</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | <b>100%</b>                 | <b>Assessment</b>           |            | <b>Total 100%</b> |
|                                        | Theoretical content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            | <b>30%</b>                  | <b>Formative assessment</b> | <b>30%</b> | <b>50%</b>        |
|                                        | Practical work:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            | <b>70%</b>                  |                             | <b>70%</b> |                   |
|                                        | ● Group project and presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>20%</b> |                             |                             |            |                   |
|                                        | ● Individual project /Work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>50%</b> |                             |                             |            |                   |
|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            | <b>Summative Assessment</b> |                             | <b>50%</b> |                   |

## Elements of Competence and Performance Criteria

|                                                  |                                                                                                        |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>1. Apply Digital Logic</b>                    | 1.1 Electronics Number systems are properly described according to their types                         |
|                                                  | 1.2 Basic digital codes are Applied according to their types                                           |
|                                                  | 1.3 Logic Gates are properly applied in line with Boolean Algebra                                      |
|                                                  | 1.4 Circuit complexity is effectively Adjusted based on its Boolean expression                         |
| <b>2. Apply Logic ICs</b>                        | 2.1 Logic Families are properly identified according to their categories                               |
|                                                  | 2.2 CMOS, TTL and ECL operations are properly applied based on their categories                        |
|                                                  | 2.3 logic ICs are properly applied based on their parameters                                           |
| <b>3. Implement Combinational Logic Circuits</b> | 3.1 Combinational Logic Circuits are clearly identified according to their types                       |
|                                                  | 3.2 Binary Arithmetic Circuits are accurately designed with reference to logic gates.                  |
|                                                  | 3.3 Multiplexer and DE multiplexer circuits are properly designed based on digital data-select methods |
|                                                  | 3.4 Encoders and decoders are correctly described based on logic circuit standards                     |
|                                                  | 3.5 Binary Comparators circuits are properly designed in line with logic gates principles.             |
| <b>4. Implement Sequential</b>                   | 4.1 Sequential Logic Circuits are properly identified based on digital logic circuits standards        |

|                       |                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Logic Circuits</b> | 4.2. Un-clocked Circuits are effectively implemented according to sequential logic circuits concepts  |
|                       | 4.3 Clocked circuit are appropriately applied based on sequential logic circuits concepts             |
|                       | 4.4 Counters and Registers are appropriately applied based on their types.                            |
|                       | 4.5 Microprocessors are appropriately described based on their types.                                 |
|                       | 4.6 Arithmetic Logic Unit is correctly Applied in accordance with sequential logic circuits concepts. |

### Intended Knowledge, Skills and Attitude

| Knowledge                                                                                                                                                                                                                                           | Skills                                                                                                                                                                                                                                                                                                                               | Attitude                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>✓ Describe digital Logic Fundamentals</li> <li>✓ Define number Systems</li> <li>✓ Define Boolean Algebra</li> <li>✓ Describe digital Components concepts</li> <li>✓ Describe electronic equipment</li> </ul> | <ul style="list-style-type: none"> <li>✓ Design Circuit diagrams</li> <li>✓ Solder /desolder electronic components</li> <li>✓ Apply Timing and Clocking</li> <li>✓ Test electronic components</li> <li>✓ Analyze electronic signals</li> <li>✓ Troubleshoot electronic circuits</li> <li>✓ Calibrate electronic equipment</li> </ul> | <ul style="list-style-type: none"> <li>✓ Integrity</li> <li>✓ Persistence</li> <li>✓ Decision Making</li> <li>✓ Curiosity</li> <li>✓ Time Management</li> <li>✓ Problem Solving</li> <li>✓ Team work</li> <li>✓ Analytical Thinking</li> <li>✓ Safety Awareness</li> <li>✓ Adaptability</li> </ul> |

## Course content

|                          |                                                                                                                                                                                                                                                                      |
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| <b>Learning outcomes</b> | <b>At the end of the module the learner will be able to:</b> <ol style="list-style-type: none"> <li>1. Apply Digital Logic</li> <li>2. Apply Logic ICs</li> <li>3. Implement Combinational Logic Circuits</li> <li>4. Implement Sequential Logic Circuits</li> </ol> |
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| Learning outcome 1: Apply Digital Logic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Learning hours: 20 |
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| Indicative content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |
| <ul style="list-style-type: none"> <li>• <b>Description of numbering system</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Types of number system base                             <ul style="list-style-type: none"> <li> Decimal</li> <li> Binary</li> <li> Octal</li> <li> Hexadecimal</li> </ul> </li> </ul> </li> <li>✓ Number system base conversion                             <ul style="list-style-type: none"> <li> Conversion from Binary system to other number systems</li> <li> Conversion from decimal system to other number systems</li> <li> Conversion from octal system to other number systems</li> <li> Conversion from Hexadecimal to other number systems</li> </ul> </li> <li>✓ Operations on number system                             <ul style="list-style-type: none"> <li> Complement of number</li> <li> Signed and unsigned numbers</li> <li> Basic laws for arithmetic operation</li> </ul> </li> </ul> |                    |

 Operations (Addition, Subtraction and Multiplication) in

 number system

- **Applying digital codes**

- ✓ Definition

- ✓ Types of digital codes

-  Binary-coded decimal code (BCD code)

-  Gray code

-  Excess-3 code

-  Bi-quinary code

-  ASCII code

-  Extended ASCII code

- ✓ Apply Types of Digital codes

- **Applying logical gates**

- ✓ Truth Table

- ✓ Types of logic gates and their symbols

-  NOT gate

-  OR gate

-  AND gate

-  NAND gate

-  NOR gate

-  XOR gate

-  XNOR gate

- ✓ Apply types of logic gates

- **Adjusting circuit complexity**

- ✓ Boolean Algebra

-  Definition



Postulates



Theorems

✓ K-map

### Resources required for the learning outcome

|                                     |                                                                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equipment</b>                    | <ul style="list-style-type: none"> <li>▪ Computer</li> <li>▪ Projector</li> <li>▪ Digital display</li> <li>▪ Analogue display</li> <li>▪ Board</li> </ul> |
| <b>Materials</b>                    | <ul style="list-style-type: none"> <li>▪ Markers/ chinks</li> <li>▪ Internet</li> </ul>                                                                   |
| <b>Tools</b>                        | <ul style="list-style-type: none"> <li>▪ Calculator</li> <li>▪ Books</li> </ul>                                                                           |
| <b>Facilitation techniques</b>      | <ul style="list-style-type: none"> <li>▪ Group discussion</li> <li>▪ Oral presentation</li> <li>▪ Practical Exercise</li> </ul>                           |
| <b>Formative assessment methods</b> | <ul style="list-style-type: none"> <li>▪ Written assessment</li> <li>▪ Practical exercise</li> </ul>                                                      |

## Indicative content

- **Identifying Logic Families**

- ✓ Definition

- ✓ Types of Logic Family



- Diode Logic (DL)



- Resistor Transistor Logic (RTL)



- Diode Transistor Logic (DTL)



- Transistor-Transistor Logic (TTL)



- Emitter Coupled Logic (ECL)



- CMOS



- BiCMOS



- Integrated Injection Logic (I<sup>2</sup>L) Family

- **Apply CMOS, TTL and ECL operations**

- ✓ Comparison between CMOS, TTL and ECL

- ✓ Guidelines of using CMOS, TTL and ECL

- ✓ Interfacing with Different Logic Families

- **Applying digital ICs**

- ✓ Definitions

- ✓ Advantages and limitations of ICs

- ✓ Interpretation of ICs specifications

- ✓ Classification of digital ICs



- Based on Technology



- Based on number of active components



- Based on applications



- Based on manufacturing method

|                                                                                                                                                                                                                                                 |                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ Applying digital ICs in electronic circuits<br> Logic gates<br> Oscillators |                                                                                                                                                                                             |
| <b>Resources required for the indicative content</b>                                                                                                                                                                                            |                                                                                                                                                                                             |
| <b>Equipment</b>                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>▪ Computer</li> <li>▪ Projector</li> <li>▪ Multimeters</li> <li>▪ DC power supply</li> <li>▪ Board</li> </ul>                                        |
| <b>Materials</b>                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>▪ Markers/ chinks</li> <li>▪ Internet</li> <li>▪ Logic ics</li> <li>▪ Breadboard</li> <li>▪ Jumper wires</li> <li>▪ Electronic components</li> </ul> |
| <b>Tools</b>                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>▪ Calculator, books</li> </ul>                                                                                                                       |
| <b>Facilitation techniques</b>                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Group discussion</li> <li>▪ Presentation</li> <li>▪ Practical exercise</li> </ul>                                                                  |
| <b>Formative assessment methods</b>                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Product assessment</li> <li>▪ Practical exercise</li> </ul>                                                                                        |

| <b>Learning outcome 3: Implement Combinational Logic Circuits</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Learning hours: 20</b> |
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| <b>Indicative content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                           |
| <ul style="list-style-type: none"> <li>● <b>Identification of combinational Logic Circuits</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Working principle</li> <li>✓ Types of combinational Logic circuit</li> <li>✓ Implementing Combinational Logic</li> </ul> </li> <li>● <b>Design Binary Arithmetic Circuits</b> <ul style="list-style-type: none"> <li>✓ Key terms <ul style="list-style-type: none"> <li> Truth table</li> <li> Circuit Simplification</li> <li> Circuit diagram</li> </ul> </li> <li>✓ Types of circuit <ul style="list-style-type: none"> <li> Half adder</li> <li> Full adder</li> <li> 4-bits binary adder</li> <li> Binary Subtractor</li> <li> 4-bits binary Subtractor</li> <li> Binary adder/ Subtractor</li> <li> 4-bits binary adder/ Subtractor</li> </ul> </li> </ul> </li> <li>● <b>Design Multiplexers and DE multiplexers circuits</b> <ul style="list-style-type: none"> <li>✓ Definition of key terms <ul style="list-style-type: none"> <li> Multiplexer</li> <li> Demultiplexer</li> </ul> </li> <li>✓ Inside multiplexer</li> </ul> </li> </ul> |                           |

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| <ul style="list-style-type: none"> <li>✓ Implementing Boolean Functions with Multiplexers</li> <li>✓ Multiplexers for Parallel-to-Serial Data Conversion</li> <li>✓ Cascading Multiplexer Circuits</li> <li>• <b>Description of Encoders and Decoders</b> <ul style="list-style-type: none"> <li>✓ Definitions</li> <li>✓ Types of encoders</li> <li>✓ Applications of encoders</li> <li>✓ Types of decoders</li> <li>✓ Application of decoders</li> <li>✓ Implementing Boolean Functions with Decoders</li> <li>✓ Cascading Decoder Circuit</li> </ul> </li> <li>• <b>Design binary Comparators circuits</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Types <ul style="list-style-type: none"> <li> Identity comparators</li> <li> Magnitude comparators</li> </ul> </li> </ul> </li> </ul> |                                                                                                                                                            |
| <b>Resources required for the indicative content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                            |
| <b>Equipment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Computer</li> <li>▪ Projector</li> <li>▪ Multimeters</li> <li>▪ DC power supply</li> <li>▪ Board</li> </ul>       |
| <b>Materials</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Markers/ chalks</li> <li>▪ Internet</li> <li>▪ Logic ics</li> <li>▪ Breadboard</li> <li>▪ Jumper wires</li> </ul> |

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|                                     | <ul style="list-style-type: none"> <li>▪ Electronic components</li> </ul>                                                  |
| <b>Tools</b>                        | <ul style="list-style-type: none"> <li>▪ Calculator</li> <li>▪ Books</li> </ul>                                            |
| <b>Facilitation techniques</b>      | <ul style="list-style-type: none"> <li>▪ Group discussion</li> <li>▪ Presentation</li> <li>▪ Practical exercise</li> </ul> |
| <b>Formative assessment methods</b> | <ul style="list-style-type: none"> <li>▪ Product assessment</li> <li>▪ Practical exercise</li> </ul>                       |

| Learning outcome 4: Implement Sequential Logic Circuits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Learning hours: 20 |
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| Indicative content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |
| <ul style="list-style-type: none"> <li>• <b>Identifying sequential Logic Circuits</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Types of Sequential Circuits</li> </ul> </li> <li>• <b>Implementing un-clocked Circuits</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Latches</li> <li>✓ Types of latches</li> </ul> </li> <li>• <b>Implementing Clocked circuits</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Multivibrators</li> <li>✓ Integrated Circuit (IC) Multivibrators</li> <li>✓ Flip-Flops</li> <li>✓ Synchronous and Asynchronous Inputs</li> <li>✓ Flip-Flop Timing Parameters</li> </ul> </li> </ul> |                    |

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| <ul style="list-style-type: none"> <li>✓ Flip-Flop Applications</li> <li>• <b>Applying Counters and Registers</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Types of counters</li> <li>✓ Modulus of a Counter</li> <li>✓ Designing Counters with Arbitrary Sequences</li> <li>✓ Types of Register</li> </ul> </li> <li>• <b>Description of Microprocessors</b> <ul style="list-style-type: none"> <li>✓ Definition</li> <li>✓ Main parts of Microprocessor <ul style="list-style-type: none"> <li> Control unit</li> <li> ALU</li> <li> Memory (Registers)</li> <li> Bus</li> </ul> </li> <li>✓ Evolution of Microprocessors</li> <li>✓ Basic Microprocessor Instructions</li> <li>✓ Addressing Modes</li> <li>✓ Microprocessor Selection</li> </ul> </li> <li>• <b>Applying Arithmetic Logic Unit</b> <ul style="list-style-type: none"> <li>✓ Arithmetic operations</li> <li>✓ Logic operations</li> </ul> </li> </ul> |                                                                                                                                                      |
| Resources required for the indicative content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                      |
| Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Computer</li> <li>▪ Projector</li> <li>▪ Multimeters</li> <li>▪ DC power supply</li> <li>▪ Board</li> </ul> |

|                                     |                                                                                                                                                                                             |
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| <b>Materials</b>                    | <ul style="list-style-type: none"> <li>▪ Markers/ chalks</li> <li>▪ Internet</li> <li>▪ Logic ics</li> <li>▪ Breadboard</li> <li>▪ Jumper wires</li> <li>▪ Electronic components</li> </ul> |
| <b>Tools</b>                        | <ul style="list-style-type: none"> <li>▪ Calculator</li> <li>▪ Books</li> </ul>                                                                                                             |
| <b>Facilitation techniques</b>      | <ul style="list-style-type: none"> <li>▪ Group discussion</li> <li>▪ Presentation</li> <li>▪ Practical exercise</li> </ul>                                                                  |
| <b>Formative assessment methods</b> | <ul style="list-style-type: none"> <li>▪ Product assessment</li> <li>▪ Practical exercise</li> </ul>                                                                                        |

## References:

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2. Whitaker, J. C. (1996) The Electronics Handbook, CRC Press (in cooperation with IEEE Press), FL, USA.
3. U.A. Bakshi & A. P Godse. (2009) Analogue and digital Electronics, Amit Residency, 412, Shanwar Peth.
4. Tomal, D. and Widmer, N. S. (2004) Electronic Troubleshooting, McGraw-Hill, USA.
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6. Theraja, B. (n.d.). *A textbook of Electrical Technology Vol IV-Electronic Devices and circuits*.
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